Impact of Sleep Deprivation on Decision Making Including Human-Computer Interaction (HCI)

Abstract

This sample academic report explores how insufficient sleep can influence attention, memory, reaction time, and decision-making. It also discusses how Human-Computer Interaction (HCI) principles can help design systems that reduce errors caused by fatigue. This document is intended as a sample project report.

Introduction

Sleep is essential for cognitive performance. Sleep deprivation reduces concentration, increases mistakes, and slows responses. HCI studies how people interact with technology and can improve interfaces for fatigued users.

Objectives

• Understand the relationship between sleep and decision-making.
• Explain HCI concepts related to usability and cognitive load.
• Discuss interface design recommendations for sleep-deprived users.

Methodology

A literature-review approach was used by examining published studies on sleep, cognition, and HCI. Illustrative examples are included for educational purposes.

Findings

Research consistently suggests that reduced sleep negatively affects attention, working memory, reaction time, and judgment. Interfaces with clear navigation, warnings, and reduced cognitive load may help users make fewer errors.

HCI Applications

Examples include fatigue alerts, simplified dashboards, adaptive interfaces, readable typography, and minimizing unnecessary notifications.

Conclusion

Adequate sleep supports better decision-making. Combining sleep science with HCI can improve user safety, productivity, and overall experience.

References

1. Matthew Walker, Why We Sleep.
2. Don Norman, The Design of Everyday Things.
3. Selected peer-reviewed literature on sleep and HCI.